

Formants

The Missing Link between Orchestration and Post-Helmholtzian Timbre Research

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A Brief History of Tim(br)e—As Seen from a Formant-Based Perspective

IN the history of Western ensemble music, timbre was addressed as an independent quality relatively late. Thus, the first written mention or definition of timbre is to be found in a French encyclopedia article about sound (*son*) written by Louis Chevalier de Jaucourt in 1765 for the well-known *Encyclopédie* of Jean Le Rond d'Alembert and Denis Diderot. Here, he described timbre as a third quality of sound (besides pitch and amplitude), which depends on the particular resonant body (de Jaucourt 1765, 346). With this description, de Jaucourt anticipates Helmholtz's definition of timbre (1863, 113): "We had seen that loudness depended on amplitude, and pitch on rapidity of vibration: nothing else was left to distinguish quality but vibrational form" (Helmholtz 1875, S. 106).¹ With the emerging interest in timbre, not only did the first orchestration treatises emerge in the eighteenth century (Roeser 1764; Francoeur 1772), but a first systematic depiction for timbre similarities was also developed for the vowels in the human voice. In his *Dissertatio de Formatione loquela* (1781), Christoph Friedrich Hellweg came up with the vowel chart (or triangle) used until today, which describes the placement of vowels based on their phonetic similarity (Hellweg 1781, 41, §57). After the term *Klangfarbe* (the German equivalent for timbre) appeared in German literature for the first time (in Weber 1822, 5), Robert Willis worked out in 1832 that fixed, emphasized frequencies in the vowel spectrum are responsible for certain vowel qualities. Hermann von Helmholtz also detected these *vowel tones* in 1863, and with the help of tuning forks he discovered that they are shaped by the resonance frequencies of the oral cavity (Helmholtz 1863, 166–181). For these *characteristic* or *vowel tones*, Ludimar Hermann introduced the term *formant* in his *Phonophotographical Studies* (of the voice) in 1894; according to Hermann, the formant describes a single emphasized and dominant partial in the vowel spectrum, which is characteristic for the vowel, and which is independent from the fundamental frequency: "Again, using the centre-of-gravity method, I tried to determine more precisely the location of the characteristic tone, or formant. I allow myself to suggest the adoption of the latter term instead of the first one, which is more cumbersome.... Furthermore, I must consider as

¹ Original quote: "Es hatte sich ergeben, dass die Stärke abhängt von der Amplitude der Schwingungen, die Tonhöhe von ihrer Anzahl; so blieb für die Unterschiede der Klangfarbe kein anderer Unterschied der Schallwellen übrig, als der ihrer Schwingungsform" (Helmholtz 1863, S. 113)."

established that the formants present in the vowel sounds must not have a harmonic relationship to the fundamental tone, but rather have a completely independent position” (Hermann 1894, 267, 272).² When Carl Stumpf, in his book *Die Sprachlaute (The Sounds of Speech)*, published in 1926, transferred these “vowel tones” or formants to the vowel chart of Hellwag (Stumpf 1926, 145) and carried out his own measurements on vowels and instrument sounds, he made two groundbreaking discoveries in the field of timbre research. First, he noticed that a formant is not a single partial but a certain frequency band (Stumpf 1926, 63). Second, he found out that timbres of musical instruments seem to have vowel-like formants too. He called them “secondary formants” (*Nebenformanten*) (Stumpf 1926, 377).

Why was this discovery so significant? Since Helmholtz’s *Lehre von den Tonempfindungen* (1863) was translated into English (*Sensations of Tone*, Helmholtz/Ellis 1875), whereas Stumpf’s *Die Sprachlaute* (1926) and other books based on Stumpf’s work were not, timbre research after Helmholtz evolved differently in German-speaking countries than in the English-speaking world. Thus, the term “formant” was only extremely rarely used in English-language literature to describe sounds produced by musical instruments, whereas in German-language literature the formant became the established term for the description of instrumental timbres through the investigations of Karl Erich Schumann. As a student of Carl Stumpf, he analyzed the sounds of wind instruments in a systematic approach by measuring all reachable pitches in different dynamics (Schumann 1929). By doing so he was able to corroborate Carl Stumpf’s results by confirming that formants of musical instruments are fixed and pitch-independent regions of the spectrum in which partials have extraordinarily strong amplitudes so that the sound impression is mainly influenced by partials located in these regions (Schumann 1929; Mertens 1975, 30–33). In 1967, David Luce and Melville Clark discovered comparable spectral areas in brass instruments sounds, which they also called “formants” (Luce and Clark 1967, 1241).

Consequently, the approaches to timbre measurement also differ fundamentally in German- and English-speaking literature: while only one or a few pitches of an instrument are usually analyzed in publications about timbre in the English-speaking literature, studies from the German-speaking literature include more frequently all reachable pitches of a musical instrument, if possible in several dynamic levels. This is because musical instruments have different characters throughout their registers, with the low register of a given instrument often sounding completely different from its middle or high register.

In 1939, long before the publication of the contemporary well-known timbre space studies (Grey 1975; Krumhansl 1989; McAdams et al. 1995; McAdams 1999, etc.), Gerhard Albersheim had already developed a first three-dimensional timbre space, the “Acoustical cylinder of sound colors” based on the principles of vowel formants and timbral brightness (Figure 13.1). In this model, instrumental timbres can be placed along three dimensions based on the following timbral features:

- Middle axis: brightness
- Perimeter (helical hue): type of vowel

² Original quote: “Wiederum habe ich nach der Schwerpunktmethod die Lage des charakteristischen Tones oder Formanten genauer zu ermitteln versucht. Ich erlaube mir den Vorschlag, die letztere Bezeichnung statt der schleppenden erstgenannten generell einzuführen ... Weiter muss ich als festgestellt ansehen, dass die in den Vokalklängen vorkommenden Formanten nicht nothwendig harmonisch zu den Grundtönen sein müssen, sondern eine ganz selbstständige Lage haben.”

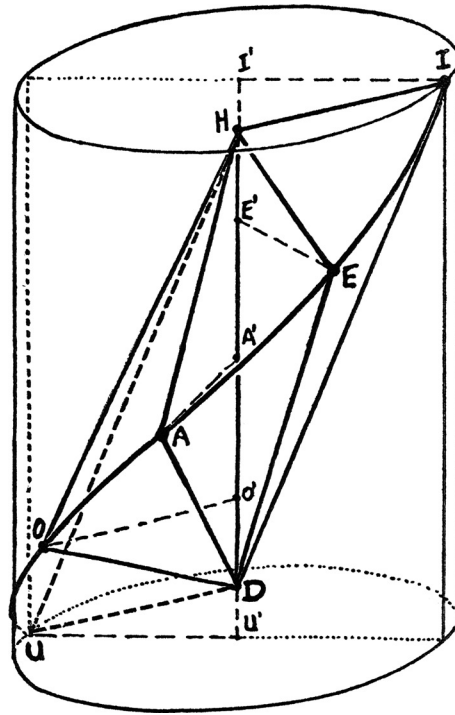


FIGURE 13.1 The Acoustical cylinder of sound colors (Albersheim 1939, 353).

- Radius: saturation (vowel-likeness, or how clearly the respective vowel formant is apparent in the spectrum)

While this first timbre space was only a pure theoretical construct, formant structures in instrument spectra have repeatedly been measured since Schumann's discovery (e.g., Reinecke 1953, 17ff.; Rösing 1970, 31–98; Burghauer and Spelda 1971, 74; Sirker 1974, 53; Dünnwald 1984, 681–682; Voigt and Heptner 1985, 525, Abbas 1989, 190–200, etc., summarized in Reuter 1995, 96–163). These and other measurements of formants were also included in Jürgen Meyer's widely used *Akustik und musikalische Aufführungspraxis* (1972, in six editions), which have been translated into English in 1999 and 2009 (*Acoustics and the Performance of Music*), so that since 1999 at the latest, measured instrumental formants can also be accessed in the English-speaking realm (like in Luce 1975, 568; Reuter 1997 and 2018; Lembke 2014; Lembke and McAdams 2012 and 2015, etc.).

How Are Formants Even Possible in Musical Instruments?

Formants characterize the timbres of non-percussive instruments like wind and string instruments. No formants have yet been detected for percussive instruments. Formants are most clearly recognizable in the spectra of pressure-controlled wind instruments such as brass and reeds (especially double reeds), whereas they are less prominent but still noticeable in flow-controlled instruments such as flutes (cf. Oehler 2008, 68ff.). Formants become recognizable when analyzing as many reachable pitches of a musical instrument as possible.

Pitch-independent fixed maxima in the acoustic spectrum over a wide range of tones are usually perceived as a stable and characteristic vowel-like quality of a given sound (e.g., bassoon sounds like “o” or oboe sounds like “æ”) and are therefore called formants. One might be tempted to trace them back to typical resonances of a musical instrument, but it is not that simple.

The spectral maxima of string instruments are mainly determined by the main air resonance of the corpus (Air mode “Ao” or Helmholtz resonance),³ the first top plate resonance (Top plate mode “T1”), and the third corpus resonance (Corpus mode “C3”).⁴ Of these resonances, Helmholtz resonance (Ao) is the most consistent and most pronounced (cf. Fletcher and Rossing 1998, 286ff.; Rossing 2010). However, compared to the pitches achievable on a string instrument, the Helmholtz resonance is very low in frequency, so that the first top plate resonance (T1) exerts a much greater timbral impact. Since the body resonances on stringed instruments are very narrow and the amplitudes of the partials on a bowed string strongly depend on the bow position, the resulting spectrum per pitch is much less clearly structured than on wind instruments. Nevertheless, according to Bynum and Rossing (2010, 254) the cello and the violin seem to have formant-like regions, respectively at about 800–1000 Hz, and 2000–3000 Hz (Curtin and Rossing 2010, 222). In other words, although one should expect fixed characteristic maxima in the spectra of string instruments due to their corpus resonances, they are much less clearly discernible than in wind instruments. This sounds paradoxical at first glance, since for wind instruments, if we consider their resonance properties according to the source-filter theory (Fant 1960), no fixed or pitch-independent formants should be expected; the tube of a wind instrument cannot be the reason why there would be a pitch-independent vowel similarity, because the resonance of a wind instrument changes with every opened or closed key, hole, or valve. From this perspective, the timbre of a wind instrument should actually change with each pitch. However, everyday experience as well as results of timbre studies described above have shown that the timbre of wind instruments remains surprisingly consistent over large registers. The explanation for the typical formants of wind instruments is therefore not to be found in the instrument’s tube resonance, but in the structure of the excitation impulses (first identified by Fransson 1966

³ The air-filled cavity in the body of string instruments acts like a Helmholtz resonator due to its two openings (f-holes). For this reason, this first prominent resonance in a string instrument’s spectrum is usually referred to as “Helmholtz resonance” in the literature, for example in Cremer 1981, 205–206: “[T]his means, as we will see, that we also have to include the velocity of the air oscillating through the f-holes into our coordinates. It represents the kinetic energy of a natural oscillation, whose potential energy is essentially contained in the compressed or expanded air present in the cavity. The associated resonance is therefore often referred to in the literature as ‘air resonance’, or even ‘air space resonance.’ However, since there are also natural oscillations at higher frequencies in the air cavity of the instrument, where it continuously oscillates, we will use the term ‘Helmholtz resonator’ for this special type of resonator, which is also commonly used in acoustic literature.” (Original quote: “[D]ies bedeutet, wie wir sehen werden, daß wir auch die Schnelle der durch die f-Löcher schwingenden Luft in unsere Koordinaten einzubeziehen haben. Sie stellt die kinetische Energie einer Eigenschwingung dar, deren potentielle Energie im Wesentlichen in der dabei im Hohlraum komprimierten oder auch expandierten Luft steckt. Die zugehörige Resonanz wird daher in der Literatur oft als „Luft-Resonanz“ oder gar „Lufttraum-Resonanz“ bezeichnet. Da es aber bei höheren Frequenzen auch Eigenschwingungen im Lufttraum des Instruments gibt, bei dem dieser als Kontinuum schwingt, wollen wir für diesen speziellen Typ den auch sonst in der akustischen Literatur gebräuchlichen Ausdruck „Helmholtz-Resonator“ verwenden.”)

⁴ The vibration modes of the top plate are abbreviated with “T,” and the vibration modes of the top plate together with the back plate with “C,” with increasing numbers according to the frequency range (Jansson, Molin, and Saldner 1994, 1101; Fletcher and Rossing 1998, 286).

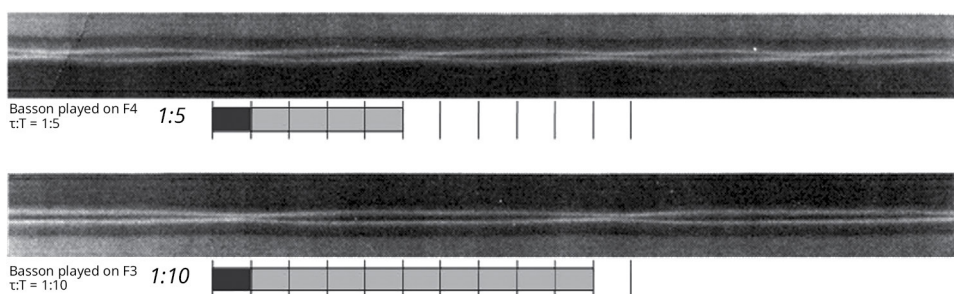


FIGURE 13.2 Continuous high-speed recording of a bassoon's double reed, filmed from the perspective of the player's mouth in an automatic blowing device, x-axis: time, y-axis: state (open vs. closed). Top: bassoon played on F4 ($\tau:T = 1:5$), bottom: bassoon played on F3 ($\tau:T = 1:10$) (based on Voigt 1975, 217, and Reuter and Oehler 2011, 70, fig. 1).

and 1967, as well as later by Fricke 1975 and Voigt 1975): by observing the opening and closing intervals of a double reed or—in the case of a brass instrument—of the lips in a mouthpiece with a high-speed camera, it becomes obvious that the duration of the closed state of the lips or reeds (as excitation pulses) remains more or less constant regardless of the pitch. These constant closing times of the lips/reeds lead to fixed, pitch-independent areas in the spectrum where partials cannot occur (spectral gaps). Between these pitch-independent spectral gaps, there are fixed spectral maxima with a formant-like structure. For the bassoon, for example, the constant duration of the closed state of the reeds is 1–1.2 milliseconds, regardless of the fundamental pitch (cf. Fransson 1967, 26; Voigt 1975, 218).

In the two high-speed recordings of oscillating bassoon reeds played on F4 (top)⁵ and F3 (bottom), as shown in Figure 13.2, it is clearly visible that the closed state is of very short and constant duration, regardless of the total period. If the duration of the closed state in double reeds is set in relation to the total period of the oscillation, then the ratio between the closed state and the total period is 1:5 for an F4, and 1:10 for an F3.

It can be seen from the spectra of the two pulse sequences that the ratio of the pulse width to total period has a decisive impact on the spectral structure: if one abstracts these two high-speed recordings into a time-amplitude representation (Figure 13.3), one can assume a pulse width τ , which forms a ratio to the total period T (i.e., $\tau:T$ corresponding to the ratio between the duration of the closed state and the total period; for F4, this would be 1:5, whereas for F3, it would be 1:10). For a $\tau:T$ ratio of 1:5, the spectral gaps (or “anti-formants”; Meyer-Eppeler 1959, 303) are located on the fifth partial and its multiples (Figure 13.3, top), while for a ratio of 1:10 they are distributed on the tenth partial and its multiples (Figure 13.3, bottom). In other words, for a pulse width of τ , the spectral minima are located in the partial intervals of $1:\tau$, regardless of the fundamental pitch. Between minima, the positions of formant-like spectral maxima evolve, as their frequencies also remain constant and independent of the fundamental pitch, due to the presence of constant spectral gaps. As a result, the sounds of bassoon and other wind instruments can convincingly be synthesized without filters, with the help of electronically generated pulse sequences with constant pulse width (τ) and variable fundamental period (T). In line with these findings the wind synthesizer “Variophon” was developed in the late 1970s based on said

⁵ The English notation is used here for the pitch description. Consequently, F4 designates the octave just above central C.

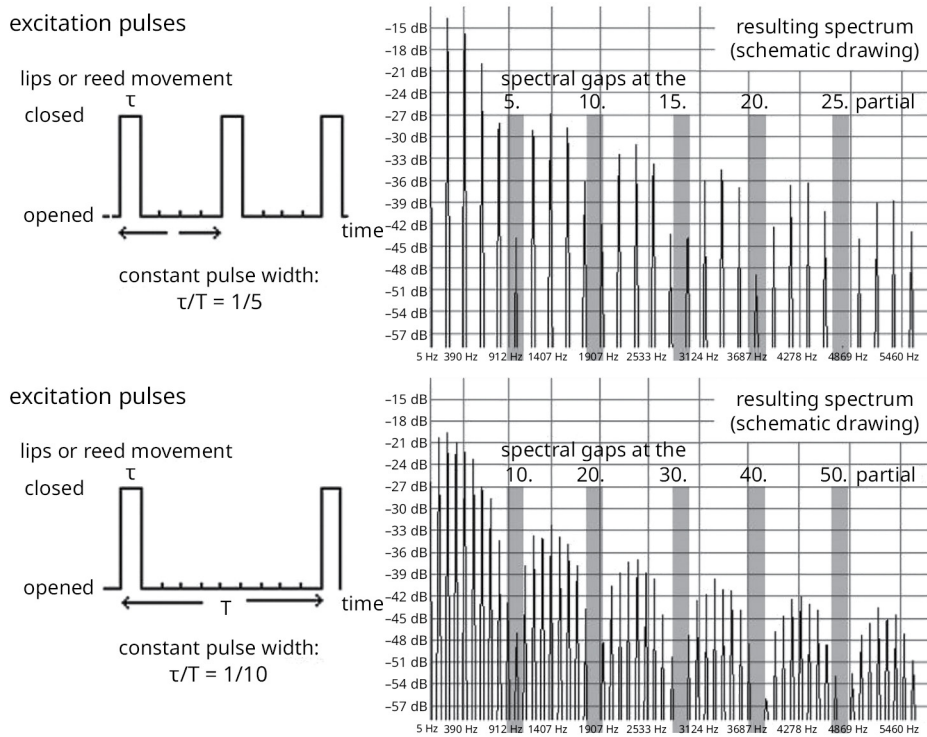


FIGURE 13.3 Abstraction of the reed vibration in a time-amplitude representation (left) and its corresponding spectrum (right). Top: for the bassoon played on F4 ($\tau/T = 1:5$), bottom: for the bassoon played on F3 ($\tau/T = 1:10$) (Oehler and Reuter 2006, 1519).

pulse-forming synthesis (Oehler 2008, 25–35). In the 1990s, this explanatory model was further refined: since the (double) reeds or lips cannot open or close infinitely fast, these processes can be even better simulated by triangular or cosine pulses than by square pulses (Auhagen 1987; Blens 1993).

How Can Formants Be Useful in Orchestration Tasks?

Formants have proved to be reliable indicators of whether a melody is heard as a whole or is split into different auditory streams based on timbre; in melodic patterns with two alternating timbres (like in perceptual experiments by Wessel 1979, 49) the formant positions can be used to predict whether one or two interleaved melodies will be audible (Reuter 2003, 214). Alternating timbres with matching first formant areas are perceived as one sole, continuous melody (one melodic stream), while alternating timbres with non-matching formant areas produce two distinct melodies in perception (two melodic streams). In that case, the melody with the higher formant in its timbre will be heard in the foreground. By shifting formant regions digitally, melodies with alternating timbres can also be specifically merged into one melodic stream or split into two distinct melodic streams. In other words, the more the formant regions of two instrument timbres overlap, the more they are perceived as sounding similar (Reuter et al. 2018).

This has serious consequences when it comes to orchestration. In the case of mixed timbres with overlapping formant areas, single instruments played in comparable dynamics and similar articulation can no longer be distinguished individually from the sound mixture, as their sounds blend homogeneously. An example of this is the case of unison combinations with horn and bassoon (especially in the lower and middle register in *mf* and *p*), or oboe and trumpet. This is what we call “timbral blend” (Reuter 1996).

On the other hand, when mixing timbres with non-overlapping formant areas, the instruments can very well be perceived individually through the sound mixture, for example in unison combinations of horn and oboe, or bassoon and trumpet (also called selective masking; see Fricke 1986).

Considering that the individual registers of musical instruments sometimes sound very different from each other, another relationship can be established between the formant range, the instrument’s register, and the degree of timbral blending. In almost all cases, the high register of an instrument starts where the fundamental reaches the first formant area. From this point on, the timbre inevitably changes (it becomes more atypical for the instrument, and more penetrating), the instrument becomes more difficult to play and can be better distinguished in ensemble playing simply because in the high register, the first formant area corresponds directly to the played fundamental and therefore changes following the pitch (Reuter 2002, 25–27).

If one summarizes the existing formant measurements on wind instruments as well as the sound-shaping resonances typical for string instruments, an overview of the pitch-independent spectral maxima of common orchestral instruments can be drawn up, as shown in Figure 13.4.

Thus, in Western music ensembles, there seem to be two particular and typical areas where the main formants of wind instruments partially or completely overlap:

- a. In the u/o [ʊ/ʊ] formant region, approximatively between 300 and 500 Hz (typical of the horn, bassoon, and tuba).
- b. In the a/æ [ʌ/æ] formant region, approximatively between 1000 and 1500 Hz—these are the regions where the flute, oboe, English horn, clarinet (in the high register), and trumpet can be found.

Instruments whose main formants are located entirely in the same region blend very homogeneously with each other (i.e., they cannot be perceived separately in ensemble playing when playing together in unison or in chords with coherent rhythmic alignment, similar articulation, and appropriate dynamics), and instruments whose main formants are located in contrasting regions only partially mask each other and can very well be perceived separately in similar ensemble playing situations. The closer the main formant areas of two instruments are located to each other, the more similar their timbres will be and the more likely timbral blend will occur.

The often very vague and experience-based recommendations made in the orchestration treatises of the last 250 years about timbral blending and timbral similarity can be summarized to the few principles described above. Thus, combinations of bassoon and French horn (both of which have their first formants between 300 and 500 Hz) are consistently recommended as particularly blending.⁶

⁶ See, e.g., Roeser 1764, 22–23; Francoeur 1772, 55–56; Albrechtsberger 1790, 180; Vandenbroeck 1793, 9; Marx 1851, 84, 145f., 148, and 347; Gleich 1853, 21; Lobe 1878, 30 and 31; Kling 1882, 33; Schubert 1885, 45; Prout 1888, 46–47 and 106; Jadassohn 1889, 242, 254, 346, and 348; Widor 1904, 46; Rimski-Korssakow 1912, 24, 57, 83, 88, and 90; Riemann 1919, 48 and 75; Korner and Rathke-Bernburger 1927; Ribate 1943, 90, etc.

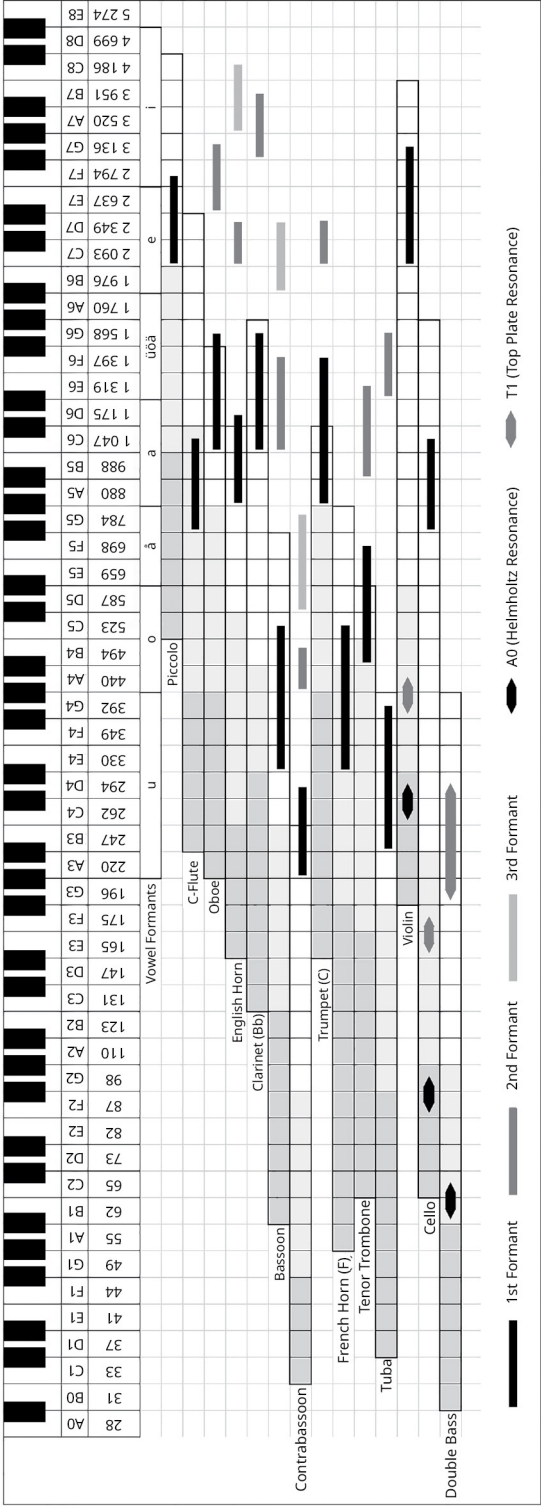


FIGURE 13.4 Compared formant areas and main resonances of various orchestral instruments (based on Reuter 2014, 401; register boundaries after Siedenburg, Jacobsen, and Reuter 2021, 3717).

The same holds true for oboe and trumpet: with their common first formant region at about 1000–1400 Hz, they blend particularly well (e.g., Marx 1851, 208, 209, 347, and 525; Rimski-Korssakow 1912, 35, 56, 88, 89, and 92; Körner and Rathke-Bernburger 1927; Ribate 1943, 90; Koechlin 1955, vol. 2, 193; Piston 1955, 427; Kunitz 1956, vol. 3, 74; Kunitz 1958, vol. 7, 555; Kunitz 1961, 20 and 55; Kennan 1962, 169; Jacob 1962, 39–40 and 65, etc.).

Conversely, oboe and French horn, with their strongly contrasting formant areas, are not recommended for timbral blending (e.g., Vandenbroeck 1793, 9; Marx 1851, 179f.; Prout 1888, 71 and 106; Jadassohn 1889, 346; Koechlin 1955, vol. 2, 187 and 235; Kunitz 1957, vol. 6, 463 and 472, etc.). And the same is true for bassoon and trumpet (e.g., Schubart 1806, 327; Koechlin 1955, vol. 2, 196 and 242; Kunitz 1958, vol. 7, 557, etc.).

However, since the oboe's main formant and the bassoon's second formant cover the same area, a homogenous timbre can be achieved by combining the bassoon's upper register with the oboe's middle register, both in unison and in octave combinations as well as in chordal ensemble playing (e.g., Jadassohn 1889, 352; Rimski-Korssakow 1912, 49 and 76; Riemann 1919, 47 and 48; Piston 1955, 423; Kunitz 1956, vol. 3, 74; Kunitz 1957, vol. 5, 287, 292, and 303; Erpf 1959, 53; Rogers 1970, 50, etc.).

All of this amounts to a great simplification of the mixing and blending recommendations for orchestral instruments from more than sixty instrumentation treatises over the past 250 years. From a formant-based perspective, they can be condensed to a few concise principles (Reuter 2002):

- Usually, if timbres with matching or strongly overlapping formant areas are mixed, the instruments can no longer be heard individually through the sound mixture; the sounds are more likely to blend homogeneously (e.g., horn/bassoon or oboe/trumpet).
- If timbres with different formant ranges are mixed, the individual instruments can be heard very distinctively as separated from the sound mixture; they only selectively mask each other (e.g., horn/oboe or bassoon/trumpet).
- The closer the first formant regions of two instruments are to each other, the more their timbres are perceived as similar and the better they blend in ensemble playing.
- If the pitch reaches the high register, most instruments' timbres undergo a clearly perceptible timbral change, since from this point onward the first formant range (which usually dominates the timbre) is now filled only by the fundamental. As a result, the instrument sounds less typical but more penetrating, and therefore the high register is usually less suitable for blending mixtures.
- If all formant regions for several musical instruments are compiled in a general overview, then two zones with coinciding or strongly overlapping main formant ranges can be identified for ensemble playing in particular:
 - A u/o [ʊ/ʊ] formant range, approximately between 300 and 500 Hz (horn, bassoon, and tuba).
 - A a/æ [ʌ/æ] formant range, approximately between 1000 and 1500 Hz (flute, oboe, English horn, clarinet in the high register, and trumpet).

Interactive and Dynamic Formant Maps as Helpful Orchestration Tools

While timbre spaces emerged in English research from the 1970s (Grey 1975) for the depiction of timbre similarities, formant maps following the same interest have been developed in

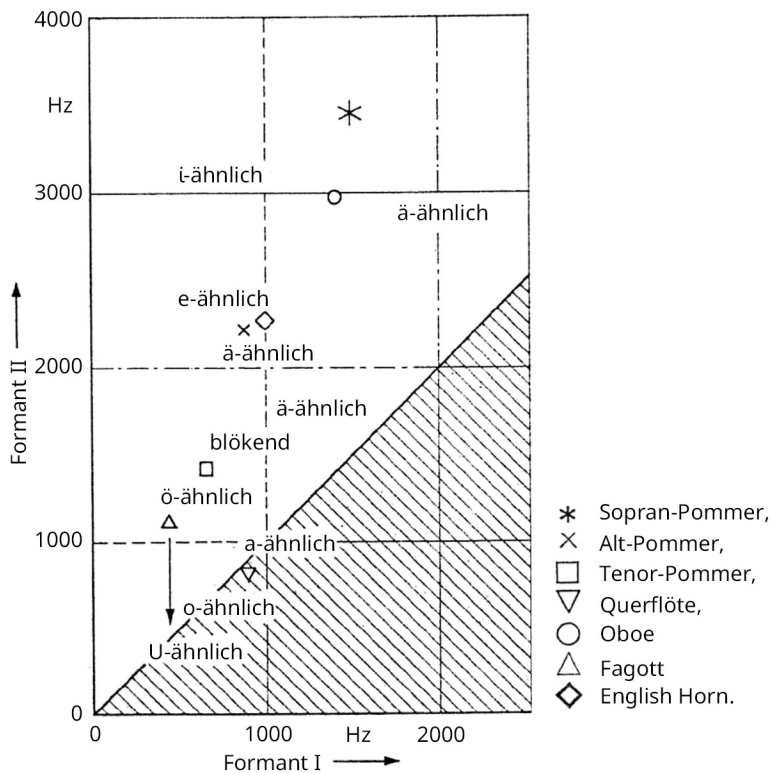


FIGURE 13.5 Formant map (‘Perceptual space’) for woodwind instruments timbres with two-formants: pommer/schalmey, flute, oboe, bassoon, English horn (Sirker 1974, 52).

German timbre research. In 1974 Udo Sirker created a first formant map named ‘Perceptual Space for Timbres with Two-Formants’ (*‘Wahrnehmungsraum bei zweiformantigen Klängen’*) based on data from the literature and from his own measurements (Figure 13.5; Sirker 1974, 52). As with the vowel charts, the *x*-axis describes the position of the first formant and the *y*-axis, the position of the second formant.

While Sirker determined the formant positions based on the strongest amplitudes in the spectrum of various instrument sounds at different pitches, the calculation of the formants in John McCarty’s ‘Vowel Space’ (2003) was done with the help of the MATLAB tool Colea (Loizou 1998).⁷ In contrast to Sirker’s ‘Perceptual Space,’ in McCarty’s ‘Vowel Space’ only one pitch was analyzed for each instrumental sound, which makes McCarty’s formant map much less generalizable. Nevertheless, McCarty formulated an observation familiar to musicians from everyday experience, namely that musical instruments often sound vowel-like, and that this timbral proximity to vowels can be explained by the vowel-like formant structure present in instrument timbres: ‘Now I can say that the tuba’s timbre has a ‘U’ sound or (oo) as in the word who, or the trumpet has an ‘æ’ timbre like the word actor’ (McCarty 2003, similar statements can already be found in Stumpf 1926, 400; Schumann 1929, 103, etc.).

⁷ Colea is a MATLAB Application for pitch and formant analysis for speech sounds developed by Philip Loizou in 1998 (<https://ecs.utdallas.edu/loizou/speech/colea.htm>, Loizou 1998).

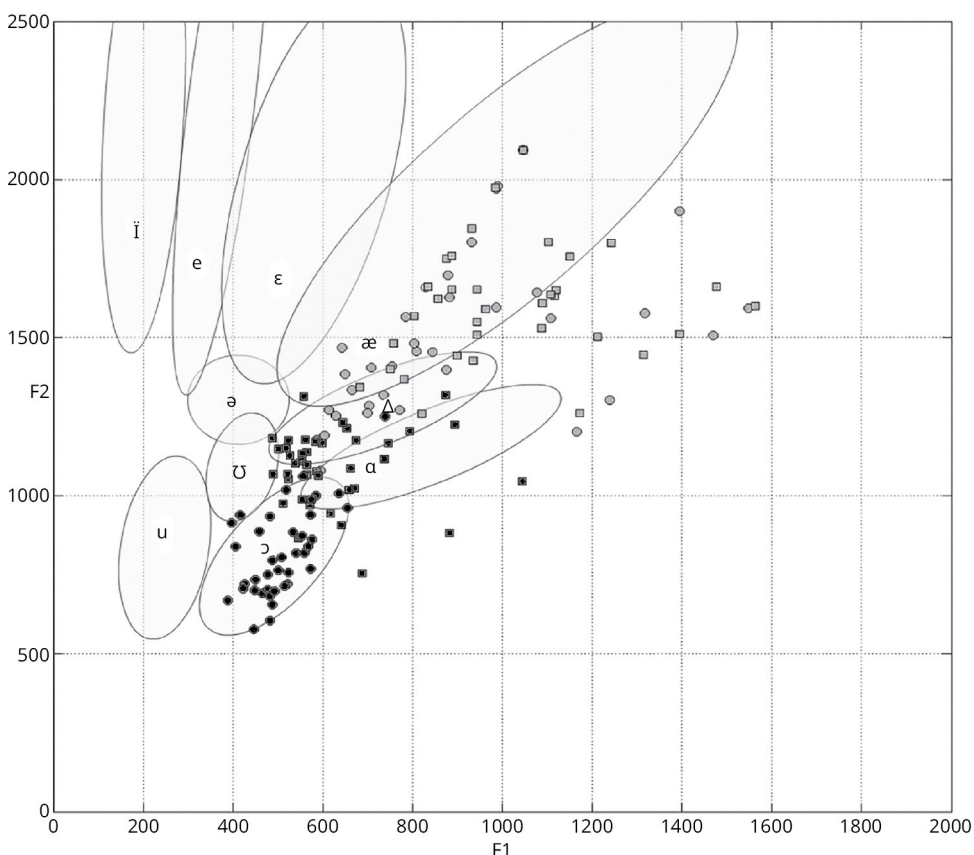


FIGURE 13.6 Formant map with the sounds of bassoon (black) and oboe (gray) in all reachable pitches in *ff* (squares) and *pp* (circles) (Reuter, Czedik-Eysenberg, Siddiq, and Oehler 2017, 191).

With the help of Praat (Boersma and Weenink 2001),⁸ Reuter et al. (2017) measured the first and second formants (F1 and F2) of conventional Western orchestral wind instruments of the Vienna Symphonic Library (VSL) in all reachable pitches and in two different dynamics (*ff* and *pp*, for a total of 593 single sounds). The results are displayed in an interactive online formant map (see an example in Figure 13.6, or follow the URL provided in the endnote).⁹ For each instrument, each combination of pitch and dynamic is symbolized by a point (for *pp* sounds) or a square (for *ff* sounds), which is placed in a 2D-graph formed by the dimensions F1 and F2 (central frequencies of formants 1 and 2). Formants of individual wind instruments can be selected arbitrarily according to register and dynamic level in order to hear the corresponding sound combination. Mean values (circle center) and standard deviation (diameter of ellipses) can also be displayed per instrument and register. It can be observed that formants are much more pronounced in

⁸ Praat is a widely used multi-purpose script-based software for phonetic signal analysis of voices and spoken sounds, developed by Paul Boersma and David Weenink since 1992 (<https://www.fon.hum.uva.nl/praat/>, Boersma, and Weenink 2001).

⁹ Available at https://muwidb.univie.ac.at/orchestration/formanten/formantregister4_english.htm (Reuter 2017).

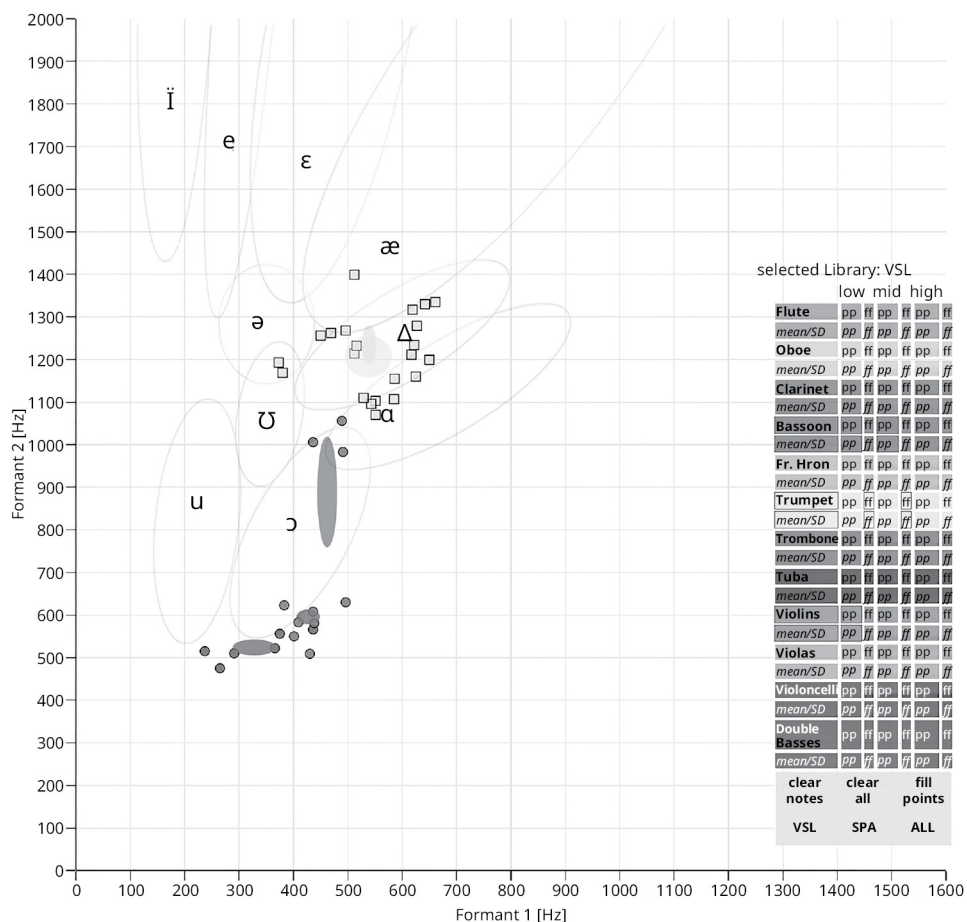


FIGURE 13.7 (From top to bottom): Formants, their mean values and standard deviations (ellipses) for trumpet (in the low and middle register, *ff*, light square dots), violins (in low register, *pp*, four dark round dots), and bassoon (in low and middle register, *pp*, slightly lighter colored round dots at the bottom of the field), as an example of timbral separability (Reuter 2020).

double reed and brass instruments than in flute and clarinet. Since in the high register the formant range is usually covered only by the fundamental (see above), fixed formant regions are mostly present in the low and middle registers.

Since the first interactive formant map was programmed in Flash, which is now deprecated, a new interactive formant map was created in JavaScript in 2020 (Figure 13.7).¹⁰ In this updated map, the formants of both wind and string instruments from the VSL were recalculated with Parselmouth (Jadoul, Thompson, and de Boer 2018).¹¹ With the integration of an additional sample library for wind instruments by Spitfire Audio, the dataset was expanded to a total of more than 1,100 evaluated individual timbres.

¹⁰ Available at <https://muwidb.univie.ac.at/orchestration/formantmap> (Reuter 2020).
¹¹ Parselmouth is a Python library adapting the Praat algorithms to a Python environment, developed by Jadoul, Thompson, and de Boer in 2018 (<https://parselmouth.readthedocs.io/en/stable/>).

While the formant concept has so far only been tested on single pitches, formant movements can also be displayed directly in a musical context, using another interactive dynamic formant map (Reuter 2021).¹² For this purpose, the formants of the individual musical instruments used in the first movement of Beethoven's Seventh Symphony were tracked via Parselmouth (Jadoul, Thompson, and de Boer 2018; the reverberation-free single tracks used for this were taken from Pätynen, Pulkki, and Lokki 2008). Via Plotly¹³ and P5,¹⁴ the collected values were synchronized with the audio files and transferred to an interactive formant field, so that the behavior of instrumental formants can be visualized "in the wild" while they are moving during the piece within specific "territories" or areas that are typical for them. All formant depictions presented here can be viewed interactively at <https://muwidb.univie.ac.at/orchestration>.

Summary and Outlook

In post-Helmholtzian timbre research, formants turned out to be a simple, intuitive, and effective way to classify timbres of musical instruments according to their vowel similarity (Reuter et al. 2024). Especially in wind instrument timbres, distinct fixed and pitch-independent characteristic spectral maxima ("formants") with spectral gaps in between can be determined. On one hand, these formant structures ensure that when musical instruments with different timbres are playing together, their formant regions are also located at different and thus particularly perceptible positions in each case, because the important partials of one instrument, which are located within its formant regions, are greatly responsible for masking the relatively unimportant partials of the other instruments, which are rather located in the spectral gaps between its formants, and vice-versa (Fricke 1986, 145). This principle of selective masking is very simple and plausible from both a timbral and a psychoacoustic point of view. According to Fricke (1993, 189), the law of the stronger applies here, and the partials located in the formant areas are the stronger ones in each case. All weaker timbral components that fall in the gaps between the formant areas are of lesser importance in terms of timbre perception and may therefore be masked by the stronger components belonging to other instruments in that spectral region.

On the other hand, there are also instrumental combinations with strongly overlapping or almost identical formant constellations; in this reverse case, the opposite of selective masking can be observed: timbral blend (Reuter 1996, 79–86). Based on these two principles (selective masking vs. timbral blending), the manifold recommendations for the mixture of wind instruments in most of the instrumentation treatises of the last 250 years can be summarized as the following (Reuter 2002, 25ff.):

- Overlapping first formants → timbres blend homogenously.
- Non-overlapping first formants → timbres are perceived separately.
- Fundamental in the first formant region → the high register is reached, and the timbre becomes very distinctive but less typical of the instrument.
- Formants of orchestral musical instruments are mostly located in different frequency bands. Nevertheless, there are two common regions (from 250 to 500 Hz as well as from

¹² Available at <https://muwidb.univie.ac.at/orchestration/formants.htm> (Reuter 2021).

¹³ Plotly JavaScript Open Source Graphing Library (available at <https://plotly.com/javascript/>).

¹⁴ P5 JavaScript library for creative coding (available at <https://p5js.org/>).

1000 to 1500 Hz), in which instruments with overlapping formant regions blend particularly well.

Furthermore, formants are very helpful for the prediction of the results in perceptual experiments about auditory grouping of melodies with alternating timbres:

- Overlapping first formants → only one melody is heard.
- Non-overlapping first formants → two interwoven melodies are heard, where the melody with the higher formant is perceived in the foreground.

Formants that have been detected on a large number of musical instrument timbres since the 1920s were especially popular for describing the timbral characteristics of musical instruments in German publications from the 1970s onward. Today, a large number of timbre features can be extracted from musical sounds by using musical signal analysis libraries such as the MIRToolbox (Lartillot and Toivianen 2007), the Timbre Toolbox (Peeters et al. 2011), Essentia (Bogdanov et al. 2013), LibROSA (McFee et al. 2015), MiningSuite (Lartillot 2019), and other ones as well. These libraries and toolboxes allow a far more accurate and multi-dimensional classification of timbres than was possible with the technical resources of the twentieth century. Considering that twentieth-century research on instrumental formants strongly contributed to a deeper understanding of how musical sounds work, it is remarkable that these intuitively applicable timbre features have not yet found their way into modern musical signal analysis libraries. One possible reason could be that formants were discovered “too early,” long before there was any idea of predictable timbre features at all. Furthermore, we have to consider that until the Second World War, scientific publications in German- and English-language literature had a similar international status. For this reason, the publications of Stumpf and Schumann were or are still not available in English, which is why they were no longer received internationally after WWII. Thus, formants tend to be associated more with phonetics (“vowels”) than with instrumental timbre.

Current algorithms for formant calculation/estimation could easily be taken from already existing phonetics libraries, like Colea (Loizou 1998), Praat (Boersma and Weenink, 2001), Parselmouth (Jadoul, Thompson, and de Boer 2018), Formant Tracker (Kamath 2005), and Formant Estimation (Rabiner et al. 2021).

An integration of formant analysis algorithms into contemporary audio signal analysis toolboxes—as one timbre dimension among many others—would certainly be a desirable enrichment of contemporary sound analysis possibilities, especially considering that there is an already existing formant research history that is almost a hundred years old.

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